# Functions

CSE 101 Section 03
Introduction to Algorithms and Computing
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## This program is one long, complex sequence of statements

```
statement
statement
statement
statement
statement
statement
statement
statement
statement
statement
statement
statement
```

## In this program, the task has been divided into smaller tasks, each of which is performed by a separate function

```
function function1()
        statement
        statement
        statement
end
function function2()
        statement
        statement
        statement
end
function function3()
        statement
        statement
        statement
end
function function4()
        statement
        statement
        statement
end
```

# What are functions?

A grouping of statements together to perform some functionality.

**Benefits**

- Readability
- Reusability
- Modularity (i.e., breaking program into smaller tasks)

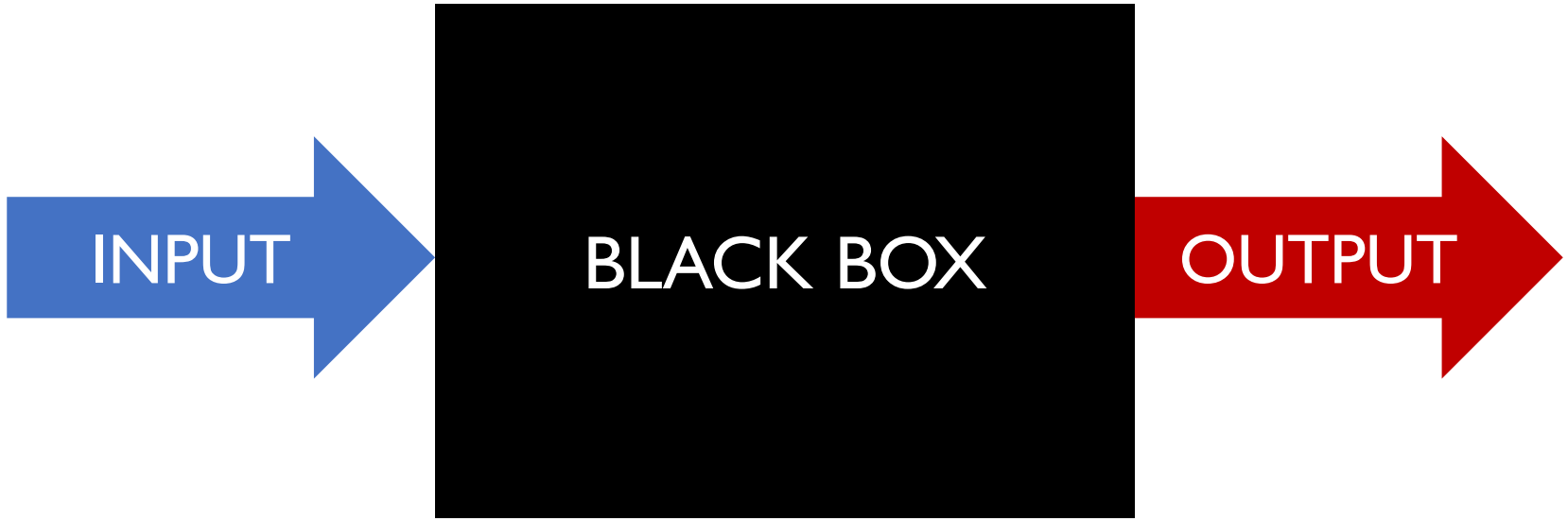
# Anatomy of a function

```
function [y1,..,yN] = myfun(x1,..,xM)
        statement(s)
end
```

The name of the function should reflect what it does. Same naming rules as for variables.

Examples: findAverage()
printReport()

# Functions



# Example

```
main()

function singHappyBD()
        disp("Happy birthday to you")
        disp("Happy birthday to you")
        disp("Happy birthday to you, Random Person")
        disp("Happy birthday to you!")
end


function main()
        singHappyBD()
        singHappyBD()
end
```

# Few rules to note about MATLAB functions

- In a Matlab script file which contains commands and function definitions, functions must be at the end of the file.

- Script files cannot have the same name as a function in the file.

- But in a function file which contains only function definitions. The name of the file must match the name of the first function in the file.

- Read more here: https://www.mathworks.com/help/matlab/ref/function.html

# Example – with an input

```
main()

function singHappyBD(person)
        disp("Happy birthday to you")
        disp("Happy birthday to you")
        disp("Happy birthday to you, " + person)
        disp("Happy birthday to you!")
end

function main()
        singHappyBD("Gracie")
        name = "Tulie";
        singHappyBD(name)
end
```

# Another Example – Not a good one!

```
distanceBetweenPoints()

function distanceBetweenPoints()
        x1 = input("Enter x1: ");
        y1 = input("Enter y1: ");
        x2 = input("Enter x2: ");
        y2 = input("Enter y2: ");
        distance = sqrt((x2-x1)^2 + (y2-y1)^2);
        disp(distance)
end
```

# Why is that not a good function?

- Not general!
- Doing more than one thing – getting input from the user, doing the calculation and printing the result on the screen.
- Requires input from the user (thus not generally reusable)
- Output just printed on the screen
  - cannot be used by the program in any way later

# More general!

```
function distance = distanceBetweenPoints(x1, x2, y1, y2)
        distance = sqrt((x2-x1)^2 + (y2-y1)^2);
end
```

- How would we get the values x1, x2, y1, y2?
  - From the user OUTSIDE the function
  - Read from an external source
  - Generated by the program somehow...
- Does only one thing – finds the distance between two points. If user input is required or the result needs to be printed, do it elsewhere!

# How does the function get the values?

```
dis = distanceBetweenPoints(20, 40, 100, 300)

function distance = distanceBetweenPoints(x1, x2, y1, y2)
        distance = sqrt((x2-x1)^2 + (y2-y1)^2);
end
```

# What can the inputs (aka arguments) look like?

```
dis = distanceBetweenPoints(20, 40, 100, 300)

dis = distanceBetweenPoints(firstx, secondx, firsty, secondy) % These need values assigned

dis = distanceBetweenPoints(getPoint(), getPoint(), getPoint(), getPoint()) % where getPoint() is a function that returns an integer

function distance = distanceBetweenPoints(x1, x2, y1, y2)
        distance = sqrt((x2-x1)^2 + (y2-y1)^2);
end
```

# Local Variables

```
function distance = distanceBetweenPoints(x1, x2, y1, y2)
        distance = sqrt((x2-x1)^2 + (y2-y1)^2);
end
```

The variable "distance" is a local variable.
It is *NOT visible anywhere outside the function*.
SO ... how do we get the result from this function?

```
dis = distanceBetweenPoints(20, 40, 100, 300)
```

# More on Local Variables

```
x = function1();
disp(x) %what is printed?
x = function2();
disp(x) %what is printed?? OOPS!

function x = function1()
        x = 5;
end

function function2()
        x = "elephant";
end

function function3()
        x = 17;
end
```

Because variables are defined inside functions, we can have many with the same names. They are NOT the same variables!!!

# More on Outputs (aka Return Values)

```
fact = factorial(5);
disp(fact);
disp(factorial(5));
result = factorial(2) + factorial(3);
disp(result);

function fact = factorial(n)
        if n==0
                fact = 1;
        else
                fact = 1;
                for i = [1:n]
                        fact = fact .* i;
                end
        end
end
```

# Functions with no output (or return value)

```
disp(singHappyBD()) % is this correct?
singHappyBD() % or is this?


function singHappyBD()
        disp("Happy birthday to you!!")
end
```

# Returning multiple values

```
[theSum, theAvg] = foo(40, 60, 50)
```

```
function [total, average] = foo(mark1, mark2, mark3)
        total = mark1 + mark2 + mark3;
        average = total/3;
end
```

# Next…

## Input / Output